Rafael Delwart

Robotics Engineer
4095 California St, San Francisco, CA 94118 — +1 (415) 802-5537 — radelwart@gmail.com

Major Projects

Wet-Dry Cycler for RNA Production
Designed and built an automated system to simulate early Earth conditions for non-enzymatic RNA synthesis, supporting origins-of-life research in collaboration with Dr. David Deamer and Dr. Milena Popovich. This project included:

- **Embedded control:** C/C++ on ESP32-S3 (ESP32) with a finite-state machine, timers/ISRs, watchdog, safe-state recovery, and JSON telemetry/logging.
- **PCB & electronics:** Custom PCB integrating DRV8825 stepper drivers and, heater switching, peltier bidirectional control, thermocouple feedback, and a 120V/24V/5V power tree; benchtop validation with DMM/scope.
- **UI:** React dashboard for live status/commands, scheduling, and log export.
- **Submodules:** Thermal plate; microfluidics-pump rehydration, vial motion stage; mixing motors; CO_2 flow; limit switches & sensors; power distribution.
- **Delivery & support:** Trained users and documented procedures; continuing maintenance and regular communication with the PhD researchers operating the systems.

3D Tracking Using an IMU for Rock Climbing
Designed a 9-DOF BNO055–based motion tracker with a minimal MCU state machine for on-wall movement analysis.

- **Embedded:** Calibration→Recording→Waiting state-machine control; BNO055 orientation/linear-accel with timestamped accel/gyro/mag data is sent over serial.
- **Data/analysis:** PC logging (CoolTerm) and MATLAB filtering/visualization; validated with gym wall trials.

Mechatronics Competition (2nd Place)
Led development of an autonomous robot for timed ball collection/deposit with robust real-time control under randomized field constraints.

- **Embedded/RT control:** C on PIC32 with interrupt-driven state machine, watchdogs, and debounced sensors.
- **Sensors/electronics:** 6 bumpers, 5 IR tape sensors; H-bridge motor control; regulated 5V (servo) and 3.3V (sensors/encoders); reverse-polarity protection with power LEDs; noise-reduced sensor wiring.

Detailed descriptions, documentation, and additional projects available at: ramfi-d.github.io/Engineering-Portfolio/

Work Experience

Blue Marble Space Institute of Science at NASA Ames, Moffett Field, CA *Jun 2026–Present*
Research Engineer — Origins-of-Life Instrumentation
Exobiology Branch, NASA Ames Research Center

Developing a wet-dry cycler for origins-of-life research in support of Dr. Milena Popovich and Dr. Alfonso Davila. Building on a robotics capstone project and the third iteration of a system already in use at UpRNA, I am responsible for design, fabrication, testing, and validation of the instrument. The device supports studies of RNA polymerization under prebiotic conditions and how genetic material may have emerged on early Earth.

UpRna, Santa Cruz, CA *Mar 2026–Present*
Research Scientist (Device Support)
Supervisor: David Deamer (Inventor of Nanopore Sequencing)

Assisted with sample preparation and analysis using Wet-dry Cyclying, TLC, Nanodrop, HPLC, and mass spec. Prototyped the wet-dry automation platform for nucleotide polymerization research and supported laboratory use by training lab users, writing SOPs, performing maintenance and iterative hardware/firmware upgrades.

UCSC, Santa Cruz, CA *Academic Year 2024–2025*
Undergraduate Tutor — ECE167: Sensors and Sensing
Supervisor: Professor Collen Josephson (cjosephson@ucsc.edu)

Led tri-weekly lab sessions for 100 plus students on sensor technologies and data processing. Covered sensing principles, calibration, signal filtering, ADC, and data acquisition. Guided students in practical lab applications using C programming and lab equipment.

Skills

Electronics: Circuit design/analysis (analog & digital); PCB fundamentals (schematic capture, layout review); power distribution systems; DMM/oscilloscope/logic analyzer.
Systems Modeling: Kinematics/dynamics; CAD (SolidWorks, Fusion 360); system architecture; feedback and controls
Programming: C, C++, Python, Verilog, MATLAB, Assembly, HTML, CSS, Git.
Fabrication: CNC machining; laser cutting; 3D printing; PCB milling; soldering and microsoldering;
Math & Physics: Signals & systems, control, probability, linear algebra; numerical modeling and error analysis.
Projects: Firmware & hardware bring-up; bench validation and log-based debugging; technical documentation.
Languages: French and English.

Education

University of California, Santa Cruz (UCSC)
Robotics Engineering, Minor in Electrical Engineering
Sept 2020 - June 2025

Dean's Honor List (4 times)
Major GPA: 3.74
Honors Graduate in both fields

References

Prof. David Deamer (deamer@soe.ucsc.edu) • **Prof. Collen Josephson** (cjosephson@ucsc.edu) • **Milena Popovich** (Milena.Popovich@nasa.gov)